

Data Source: [divvy\\_tripdata](#) [accessed on 04/03/23]

```
#installing packages
```

```
install.packages("tidyverse")  
library(tidyverse)
```

```
install.packages("modeest")  
library(modeest)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(lubridate)
```

```
library(readcsv)
```

```
#collecting information
```

```
year2019 <- read_csv("Divvy_Trips_2019_Q1.csv")
```

```
year2020 <- read_csv("Divvy_Trips_2020_Q1.csv")
```

```
##transformation of start/end station mutation to numeric format
```

```
year2019 <- mutate(year2019, from_station_id=as.numeric(from_station_id),  
to_station_id=as.numeric(to_station_id))
```

```
year2020 <- mutate(year2020, start_station_id=as.numeric(start_station_id),  
end_station_id=as.numeric(end_station_id))
```

```
#?transformation to date format
```

```
year2019 <- mutate(year2019, start_time=as.Date(start_time), end_time=as.Date(end_time))
```

```
year2020 <- mutate(year2020, started_at=as.Date(started_at), ended_at=as.Date(ended_at))
```

```
#adding day, week and month
```

```
year2019$d <- as.Date(year_2019$start_time)
```

```
year2019$m <- format(as.Date(year_2019$d), "%m" )
```

```
year2019$d <- format(as.Date(year_2019$d), "%d")
```

```
year_2019$day_of_week <- format(as.Date(year_2019$d), "%a")
```

```
#removing unnecessary columns
```

```
year_2019 <- year_2019 %>% select(-c(tripduration, bikeid, from_station_id, from_station_name,  
to_station_id, to_station_name))
```

```
#trip duration
```

```
year2019$trip_duration <- difftime(year2019$end_time, year2019$start_time)
```

```
#dropping NA lines
```

```
year2019_na <- drop_na(year2019)
```

```
#number of dropped rows
```

```
365069-345357=19712
```

#dropping negative values

```
year2019_d_n <- year2019_d[year2019_d$tripduration>0,]
```

```
nrows=345357
```

#using function aggregate to find difference between subscribers and customers in mins

```
aggregate(year2019_d_n$tripduration/60 ~ year2019_d_n$usertype, FUN=mean)
```

|   |          |          |
|---|----------|----------|
| 1 | Customer | 37.02064 |
|---|----------|----------|

|   |            |          |
|---|------------|----------|
| 2 | Subscriber | 13.85694 |
|---|------------|----------|

```
aggregate(year2019_d_n$weekday~year2019_d_n$usertype, FUN=mfv)
```

#using statistical model to see which days are most popular

|   |          |     |
|---|----------|-----|
| 1 | Customer | Sat |
|---|----------|-----|

|   |            |     |
|---|------------|-----|
| 2 | Subscriber | Thu |
|---|------------|-----|

```
aggregate(year2019_d_n$tripduration/60 ~ year2019_d_n$member_casual + year2020$wid,  
FUN=mean)
```

```
year2019_d_n$usertype year2019_d_n$wd year2019_d_n$tripduration
```

|   |          |     |           |
|---|----------|-----|-----------|
| 1 | Customer | Fri | 2291.2963 |
|---|----------|-----|-----------|

|   |            |     |          |
|---|------------|-----|----------|
| 2 | Subscriber | Fri | 831.5879 |
|---|------------|-----|----------|

|   |          |     |           |
|---|----------|-----|-----------|
| 3 | Customer | Mon | 1981.8979 |
|---|----------|-----|-----------|

|   |            |     |          |
|---|------------|-----|----------|
| 4 | Subscriber | Mon | 876.5452 |
|---|------------|-----|----------|

|   |          |     |           |
|---|----------|-----|-----------|
| 5 | Customer | Sat | 2304.1052 |
|---|----------|-----|-----------|

|   |            |     |           |
|---|------------|-----|-----------|
| 6 | Subscriber | Sat | 1017.5365 |
|---|------------|-----|-----------|

|   |          |     |           |
|---|----------|-----|-----------|
| 7 | Customer | Sun | 2597.7100 |
|---|----------|-----|-----------|

|   |            |     |           |
|---|------------|-----|-----------|
| 8 | Subscriber | Sun | 1002.4099 |
|---|------------|-----|-----------|

|   |          |     |           |
|---|----------|-----|-----------|
| 9 | Customer | Thu | 2436.3763 |
|---|----------|-----|-----------|

|    |            |     |           |
|----|------------|-----|-----------|
| 10 | Subscriber | Thu | 718.3761  |
| 11 | Customer   | Tue | 1889.0932 |
| 12 | Subscriber | Tue | 861.3484  |
| 13 | Customer   | Wed | 1726.3519 |
| 14 | Subscriber | Wed | 722.7948  |

#exploring features

```
aggregate(year2019_d_n$wd~year2019_d_n$usertype, FUN=mfv)
```

```
year2019_d_n$wd <- ordered(year2019_d_n$wd, levels=c("Sunday", "Monday",
"Tuesday", "Wednesday", "Thursday", "Friday", "Saturday" ))
```

```
aggregate(year2019_d_n$tripduration ~ year2019_d_n$usertype + year2019_d_n$wd, FUN=mean)
```

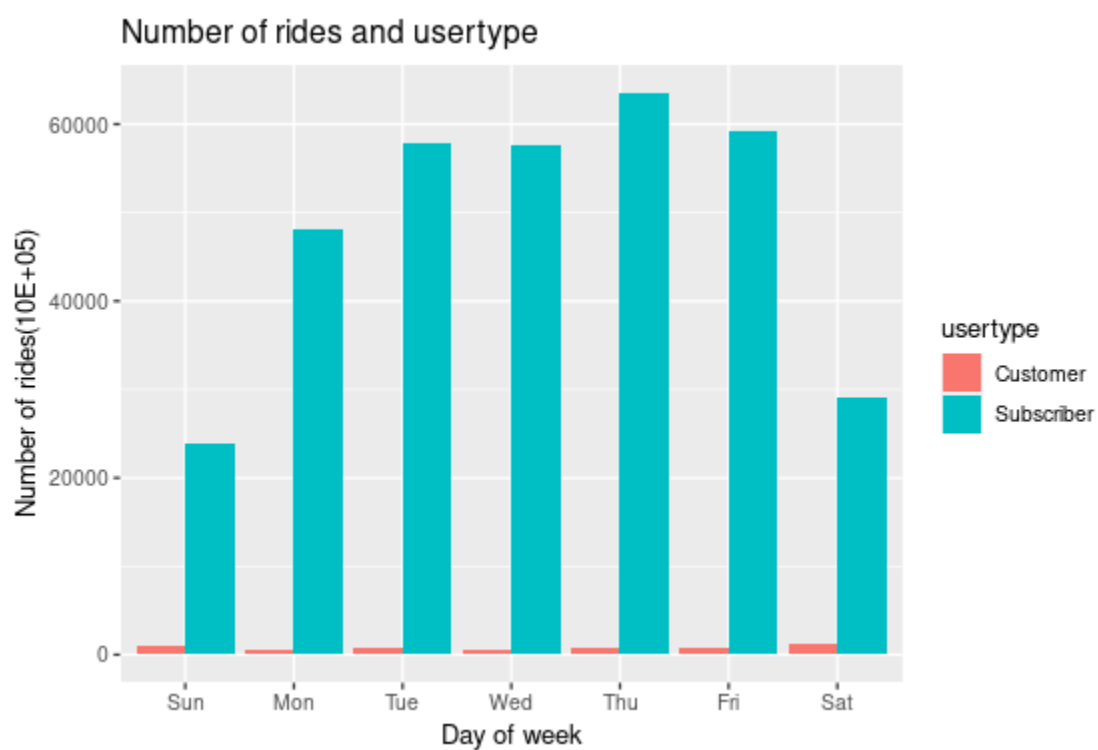
```
year2019_d_n %>% mutate(weekday=wday(start_time, label=TRUE)) %>% group_by(usertype,
weekday) %>% summarise(number_of_rides=n(), average_dur=mean(tripduration)) %>%
arrange(usertype, weekday)
```

| usertype | weekday | number_of_rides | average_dur |
|----------|---------|-----------------|-------------|
|----------|---------|-----------------|-------------|

| <chr> | <ord> | <int> | <dbl> |
|-------|-------|-------|-------|
|-------|-------|-------|-------|

|   |            |     |       |       |
|---|------------|-----|-------|-------|
| 1 | Customer   | Sun | 976   | 2598. |
| 2 | Customer   | Mon | 568   | 1982. |
| 3 | Customer   | Tue | 783   | 1889. |
| 4 | Customer   | Wed | 628   | 1726. |
| 5 | Customer   | Thu | 784   | 2436. |
| 6 | Customer   | Fri | 874   | 2291. |
| 7 | Customer   | Sat | 1321  | 2304. |
| 8 | Subscriber | Sun | 23988 | 1002. |
| 9 | Subscriber | Mon | 48161 | 877.  |

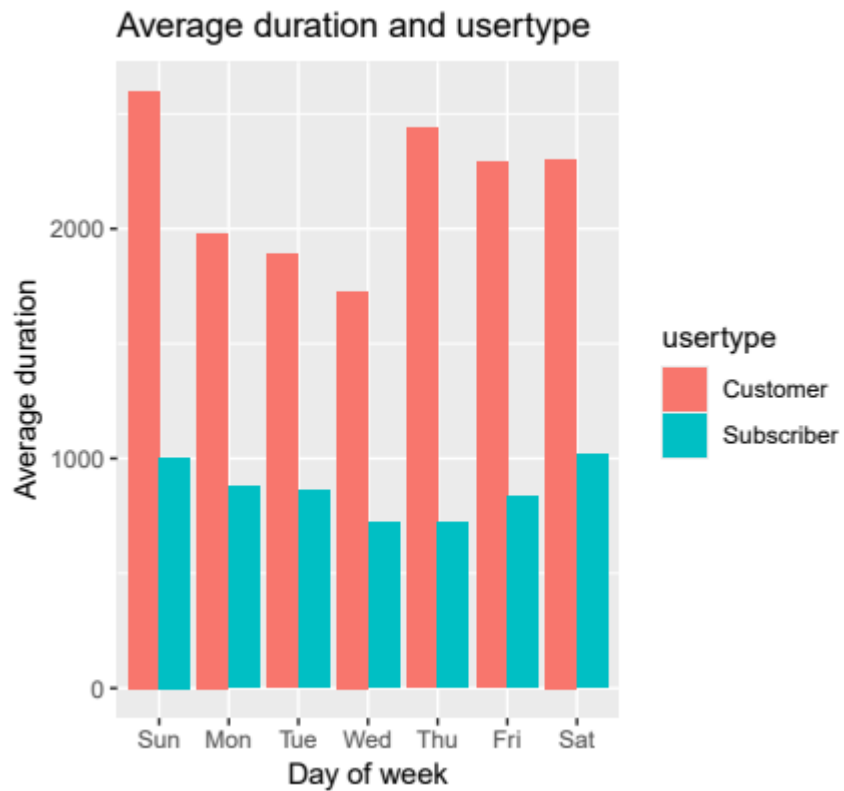
|                   |       |       |
|-------------------|-------|-------|
| 10 Subscriber Tue | 57928 | 861.  |
| 11 Subscriber Wed | 57521 | 723.  |
| 12 Subscriber Thu | 63519 | 718.  |
| 13 Subscriber Fri | 59244 | 832.  |
| 14 Subscriber Sat | 29062 | 1018. |



## Average duration

```
year2019_d_n %>% mutate(weekd=wday(start_time, label=TRUE)) %>% group_by(usertype,
weekd) %>% summarise(number_of_rides=n(), average_dur=mean(tripduration)) %>%
arrange(usertype, weekd) %>%
```

```
ggplot(aes(x=weekd, y=average_dur, fill=usertype)) + geom_col(position="dodge") +
labs(title="Average duration and usertype") + ylab("Average duration") + xlab("Day of week")
```

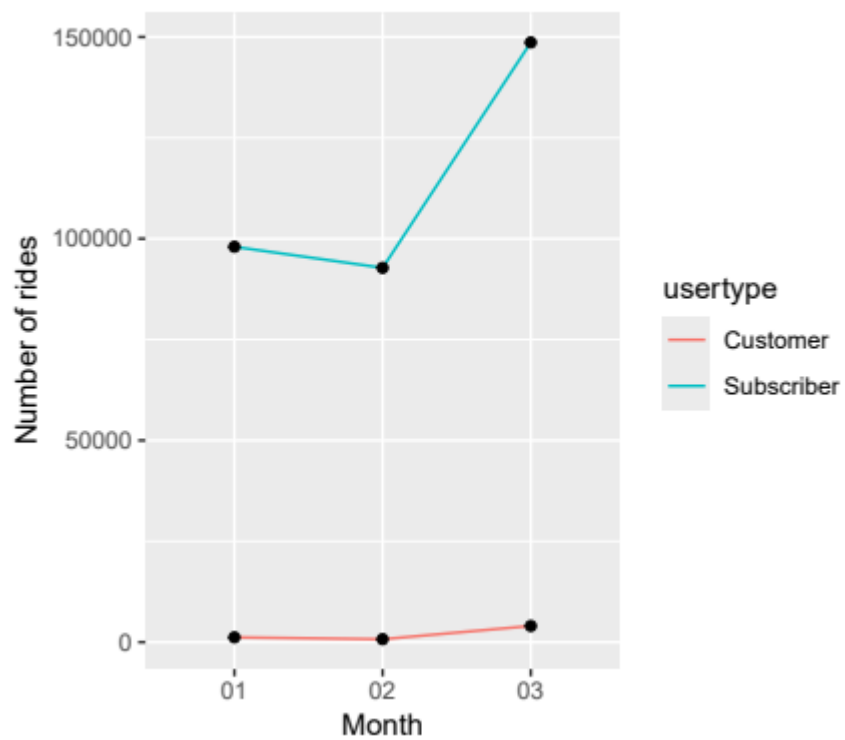


```
## Number of rides by month and usertype
```

```
year2019_d_n %>% mutate(weekd=wday(start_time, label=TRUE)) %>% group_by(usertype,
weekd) %>% summarise(number_of_rides=n(), average_dur=mean(tripduration)) %>%
arrange(usertype, weekd) %>%
```

```
ggplot(aes(x=weekd, y=average_dur, fill=usertype)) + geom_col(position="dodge") +
labs(title="Average duration and usertype") + ylab("Average duration") + xlab("Day of week")
```

Number of rides by month and usertype



```
# year2020 initialising data
```

```
year2020 <- read_csv("Divvy_Trips_2020_Q1.csv")
```

```
# changing format of data
```

```
year2020 <- mutate(year2020, start_station_id=as.numeric(start_station_id),  
end_station_id=as.numeric(end_station_id))
```

```
# adding new columns for date, day of week and month
```

```
year2020$d <- as.Date(year2020$started_at)
```

```
year2020$m <- format(as.Date(year2020$d), "%m" )
```

```
year2020$day <- format(as.Date(year2020$d), "%d")
```

```
year2020$wd <- format(as.Date(year2020$d), "%a")
```

```
# removing unnecessary columns
```

```
year2020 <- year2020 %>% select(-c(start_lat, start_lng, end_lat, end_lng))
```

```
# adding column tripduration and calculating length of a trip
```

```
year2020 <- read_csv("Divvy_Trips_2020_Q1.csv")
```

```
# removing NA rows
```

```
year2020 <- read_csv("Divvy_Trips_2020_Q1.csv")
```

```
dropped 1 row
```



```
# removing negative rows
```

```
year2020 <- year2020_d[year2020_d$tripduration>0,]
```

```
# look at summary
```

```
summary(year2020)
```

```
# look at time in minutes
```

```
summary(year2020$tripduration)/60
```

```
aggregate(year2020$tripduration/60 ~ year2020$member_casual, FUN=mean)
```

```
year2020$member_casual year2020$tripduration/60
```

```
1      casual      96.20008
```

```
2      member     12.67711
```

```
aggregate(year2020$tripduration/60 ~ year2020$member_casual, FUN=median)
```

```
year2020$member_casual year2020$tripduration/60
```

```
1      casual     21.28333
```

```
2      member     8.58333
```

```
# we can see which day is most popular
```

```
aggregate(year2020$wd ~ year2020$member_casual, FUN=mfv)
```

```
1      casual    Sun
```

```
2      member    Tue
```

# rides dependent of day of week

```
aggregate(year2020$tripduration/60 ~ year2020$member_casual + year2020$wid, FUN=mean)
```

```
year2020$member_casual year2020$wid year2020$tripduration/60
```

|    |        |     |           |
|----|--------|-----|-----------|
| 1  | casual | Fri | 119.09372 |
| 2  | member | Fri | 12.62207  |
| 3  | casual | Mon | 74.71312  |
| 4  | member | Mon | 12.97714  |
| 5  | casual | Sat | 100.28593 |
| 6  | member | Sat | 15.49982  |
| 7  | casual | Sun | 95.17611  |
| 8  | member | Sun | 15.82234  |
| 9  | casual | Thu | 127.00249 |
| 10 | member | Thu | 11.55388  |
| 11 | casual | Tue | 84.95958  |
| 12 | member | Tue | 11.53387  |
| 13 | casual | Wed | 75.74903  |
| 14 | member | Wed | 11.65894  |

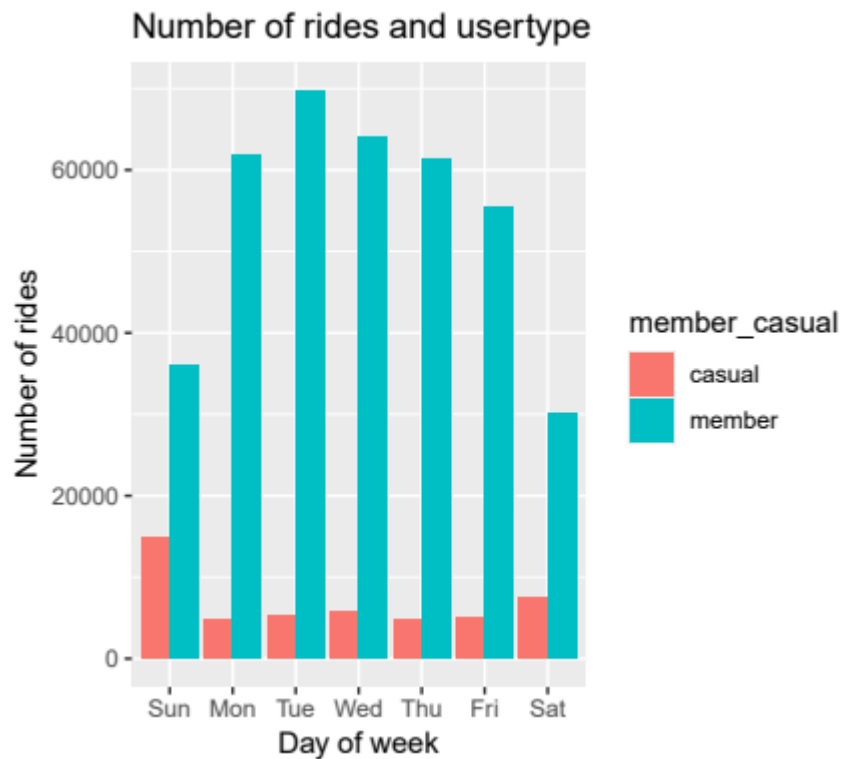
# relations

```
year2020 %>% mutate(weekday=wday(started_at, label=TRUE)) %>% group_by(member_casual,  
weekday) %>% summarise(number_of_rides=n(), average_dur=mean(tripduration/60)) %>%  
arrange(member_casual, weekday)
```

| member_casual | weekday | number_of_rides | average_dur |
|---------------|---------|-----------------|-------------|
| 1 casual      | Sun     | 14886           | 95.2        |
| 2 casual      | Mon     | 4802            | 74.7        |
| 3 casual      | Tue     | 5244            | 85.0        |
| 4 casual      | Wed     | 5874            | 75.7        |
| 5 casual      | Thu     | 4851            | 127.        |
| 6 casual      | Fri     | 5133            | 119.        |
| 7 casual      | Sat     | 7480            | 100.        |
| 8 member      | Sun     | 35964           | 15.8        |
| 9 member      | Mon     | 61923           | 13.0        |
| 10 member     | Tue     | 69697           | 11.5        |
| 11 member     | Wed     | 63978           | 11.7        |
| 12 member     | Thu     | 61245           | 11.6        |
| 13 member     | Fri     | 55496           | 12.6        |
| 14 member     | Sat     | 30104           | 15.5        |

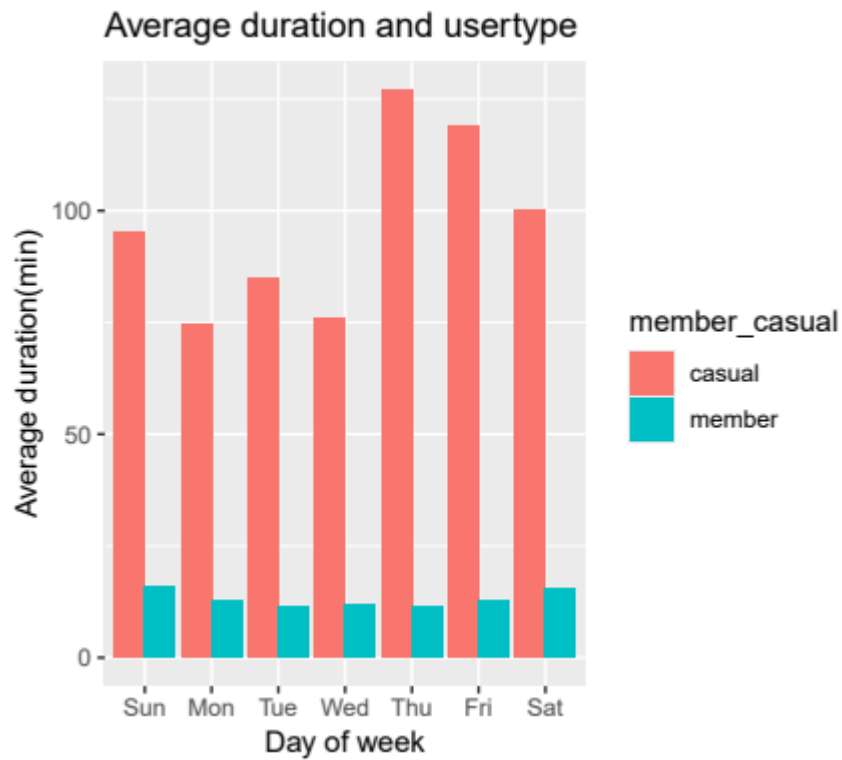
# Number of rides and day of week

```
year2020 %>% mutate(weekday=wday(started_at, label=TRUE)) %>% group_by(member_casual,
weekday) %>% summarise(number_of_rides=n(), average_duration=mean(tripduration/60)) %>%
  arrange(member_casual, weekday) %>% ggplot(aes(x=weekday, y=number_of_rides,
fill=member_casual)) + geom_col(position="dodge") + labs(title = "Number of rides and usertype") +
  xlab("Day of week") + ylab("Number of rides")
```



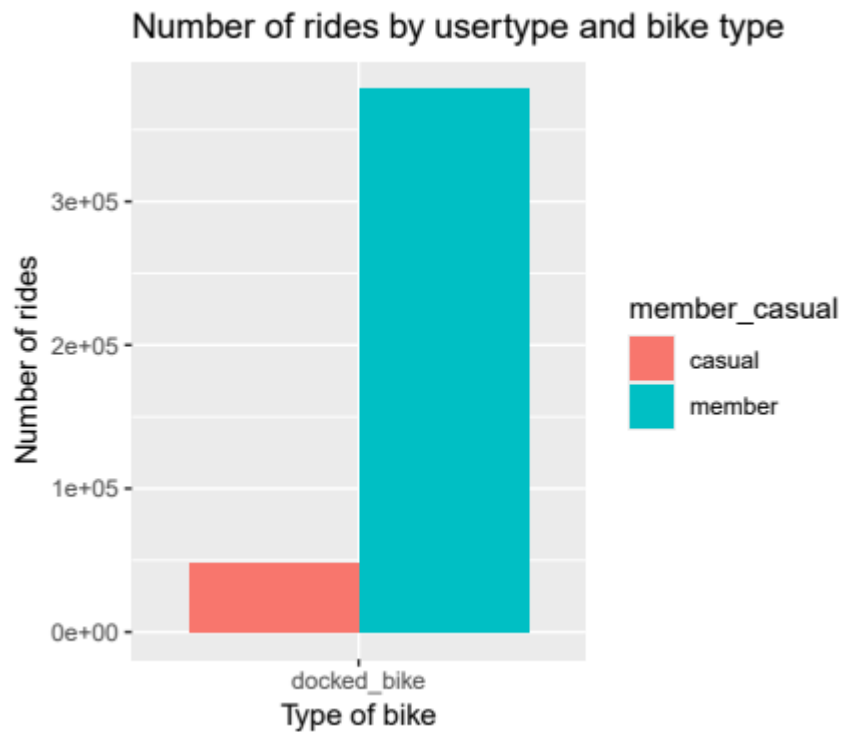
# Average duration and day of week

```
year2020 %>% mutate(weekday=wday(started_at, label=TRUE)) %>% group_by(member_casual,
weekday) %>% summarise(number_of_rides=n(), average_duration=mean(tripduration/60)) %>%
  arrange(member_casual, weekday) %>% ggplot(aes(x=weekday, y=average_duration,
fill=member_casual)) + geom_col(position="dodge") + labs(title = "Average duration and usertype") +
  xlab("Day of week") + ylab("Average duration(min)")
```



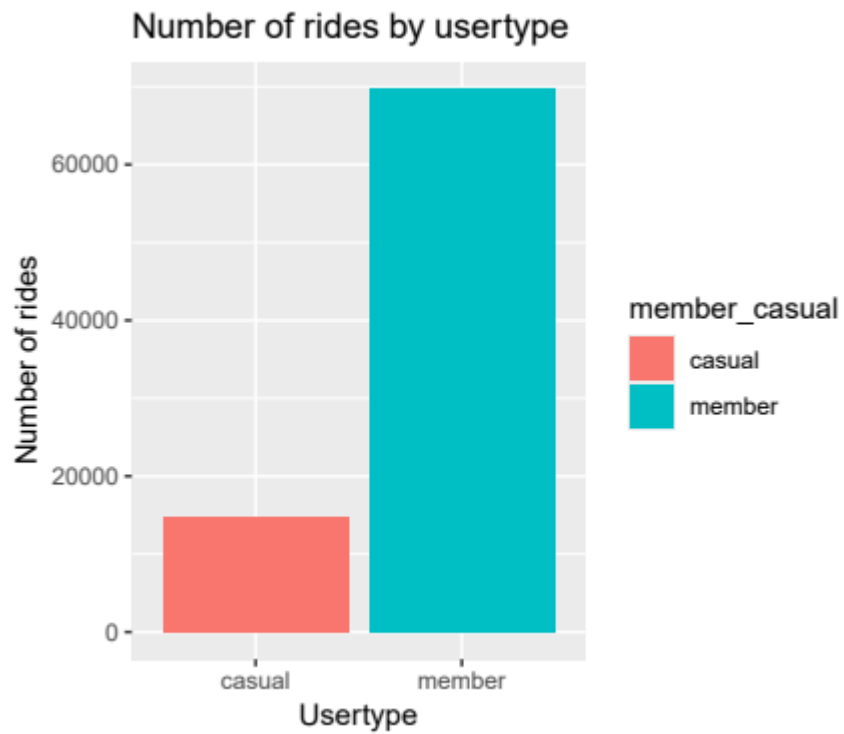
# Number of rides by bike type and usertype

```
year2020 %>% mutate(weekday=wday(started_at, label=TRUE)) %>% group_by(member_casual,
rideable_type) %>% summarise(number_of_rides=n(), average_duration=mean(tripduration/60)) %>%
  arrange(member_casual, rideable_type) %>% ggplot(aes(x=rideable_type, y=number_of_rides,
fill=member_casual)) + geom_col(position="dodge") + labs(title = "Number of rides by usertype and
bike type") +
  xlab("Type of bike") + ylab("Number of rides")
```



#Number of rides by usertype

```
year2020 %>% mutate(weekday=wday(started_at, label=TRUE)) %>% group_by(member_casual,
weekday) %>% summarise(number_of_rides=n(), average_duration=mean(tripduration/60)) %>%
  arrange(member_casual, weekday) %>% ggplot(aes(x=member_casual, y=number_of_rides,
fill=member_casual)) + geom_col(position="dodge") + labs(title = "Number of rides by usertype") +
  xlab("Usertype") + ylab("Number of rides")
```



# Number of rides by month and usertype

```
year2020 %>% group_by(member_casual, m) %>% summarise(number_of_rides=n(),  
average_duration=mean(tripduration/60)) %>%  
  arrange(member_casual, m) %>% ggplot(aes(x=m, y=number_of_rides, group=member_casual)) +  
  geom_line(aes(color=member_casual)) + labs(title = "Number of rides by month and usertype") +  
  xlab("Month") + ylab("Number of rides")
```

Number of rides by month and usertype

